Pakistan Navy Complaint Management System (CMS)
Document 02: Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) provides a comprehensive, rigorous technical description of the **Pakistan Navy Complaint Management System (CMS)**. It defines all functional capabilities, performance criteria, security parameters, database interactions, external API contracts, and structural interfaces required by the system.

1.2 Document Conventions

- **Keywords:** The terms MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, and RECOMMENDED are interpreted according to RFC 2119.
- **Identifiers:** Functional requirements are prefixed with `FR-`, Non-Functional with `NFR-`, and Interface requirements with `IFR-`.

1.3 Intended Audience

This document is prepared for:
1. **Software Engineering & DevOps Teams:** For maintenance, future module development, and deployment orchestration.
2. **Quality Assurance (QA) & Testing Teams:** As the primary baseline for unit, functional, integration, and regression test suites.
3. **Naval IT & Cyber Security Auditors:** To verify compliance with data security, authorization boundaries, and encryption standards.
4. **Project Managers & Naval Command:** To validate deliverables against contracted operational requirements.

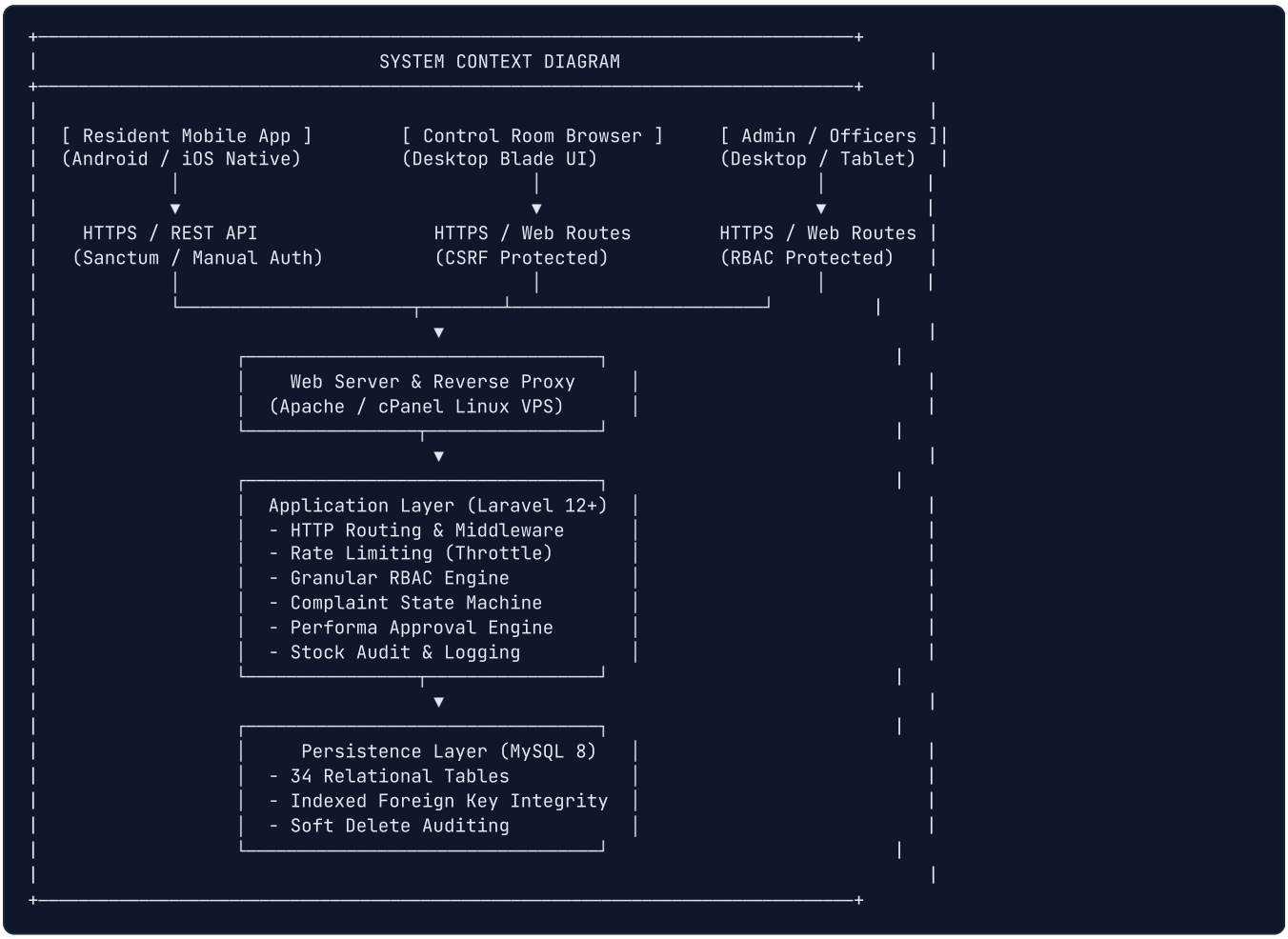
1.4 Project Scope

The CMS is an enterprise web and API platform that manages civic, infrastructure, and technical complaint handling for Pakistan Navy housing directorates and CMES formations. It encompasses:
* Web-based administrative and operational portal.
* Mobile REST API gateway for resident smartphones and field devices.
* Inventory tracking and multi-tier electronic approval performa workflows.
* Granular Role-Based Access Control (RBAC) across 19 system modules.

2. Overall Description

2.1 Product Perspective & System Architecture

The CMS operates as a three-tier decoupled client-server architecture:



2.2 Operating Environment

- **Server Operating System:** Linux (CentOS / CloudLinux via cPanel VPS)
- **PHP Runtime:** PHP 8.2 (specifically `/opt/cpanel/ea-php82/root/usr/bin/php`)
- **Web Server:** Apache 2.4 with `mod_rewrite`, `mod_headers`, and custom security `.htaccess`
- **Database Engine:** MySQL 8.0 Community / MariaDB 10.6+ (InnoDB Engine)
- **Client Browsers:** Google Chrome 90+, Mozilla Firefox 88+, Microsoft Edge 90+, Safari 14+
- **Mobile Operating Systems:** Android 8.0+ and iOS 13.0+

2.3 User Classes and Access Hierarchy

1. **Super Administrator (Role ID 1):** Unrestricted, immutable master access to all platform configurations, routes, audits, and database records.
2. **Executive Engineer / CMES Staff:** High-level supervisory review, budget approvals, and complex performa authorization.

3. **Garrison Engineer (GE) / Maintenance Supervisor:** Sector-level operational supervisory access, technician allocation, and performa authorization.
4. **Helpdesk / Data Entry Operator:** Front-office complaint registration, resident lookups, and basic technician dispatching.
5. **Field Technician / Employee:** Read-only access to assigned complaints, material requests, and progress updates.
6. **Resident / House Occupant:** Mobile app authenticated user restricted strictly to self-registered complaints, profile settings, and satisfaction feedback.

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3. Detailed Functional Requirements

3.1 Module 1: Authentication & Session Management (FR-AUTH)

- **FR-AUTH-001 (Multi-Guard Authentication):** The system MUST support independent authentication guards for administrative personnel (`web` guard via `users` table) and housing occupants (`api` / `manual.auth` guard via `houses` and `frontend_users` tables).
- **FR-AUTH-002 (Password Encryption):** All user credentials MUST be hashed using the bcrypt algorithm with a minimum work factor of 10. Plaintext passwords MUST NEVER be logged or persisted.
- **FR-AUTH-003 (Device Binding Verification):** Mobile API requests MUST pass device verification against the `registered_devices` registry, confirming IMEI/UUID status before granting access tokens.
- **FR-AUTH-004 (Audit Trail Logging):** Every administrative and resident authentication event (success or failure) MUST be recorded in the `login_history` table with user ID, IP address, user-agent string, and timestamp.
- **FR-AUTH-005 (Rate Limiting Protection):** The authentication endpoints MUST enforce strict throttling: maximum 5 login attempts per minute per IP address (`throttle:5,1`).

3.2 Module 2: Granular Role-Based Access Control (FR-RBAC)

- **FR-RBAC-001 (Isolated Module Permissions):** The authorization layer MUST evaluate exact-match module permission strings without automatic cascading or parent-to-child inheritance.
- **FR-RBAC-002 (Supported Modules):** The system MUST maintain isolated permission definitions for the following 19 modules:
  `dashboard` , `users` , `frontend-users` , `cmes` , `city` , `sector` , `roles` , `employees` , `designation` , `houses` , `complaints` , `category` , `sub-category` , `complaint-titles` , `approvals` , `spares` , `reports` , `sla` , `registered-devices` .
- **FR-RBAC-003 (Route Middleware Enforcement):** Every protected administrative route MUST be guarded by the `permission:{module}.{action}` middleware.
- **FR-RBAC-004 (Superadmin Auto-Assignment):** Upon role creation or update, if `role_id === 1` , all 19 module permissions MUST be automatically synchronized via Eloquent model lifecycle hooks (`Role::boot()`).

3.3 Module 3: Geographic & Structural Directory (FR-GEO)

- **FR-GEO-001 (City Management):** System MUST allow administrative CRUD operations for regional cities.
- **FR-GEO-002 (CMES Formation):** System MUST map Commander Military Engineering Services (CME) units to respective cities.
- **FR-GEO-003 (Sector Partitioning):** System MUST support sector definitions linked to specific CME units and cities.

- **FR-GEO-004 (House Registry):** System MUST maintain records of individual houses/accommodations, capturing Sector, House Number, Block, Category, Contact Number, and Occupant Credentials.

3.4 Module 4: Staff & Technician Roster (FR-STAFF)

- **FR-STAFF-001 (Designations):** System MUST manage job titles/trades (e.g., Electrician, Line-man, Plumber, Valve-man, Carpenter, Mason).
- **FR-STAFF-002 (Employee Profiles):** System MUST record technician details, CNIC, mobile numbers, active status, and designated trade/designation.
- **FR-STAFF-003 (Workload Assignment):** The dispatch interface MUST display active complaints assigned per technician to prevent unbalanced allocations.

3.5 Module 5: Complaint Lifecycle Engine (FR-TICK)

- **FR-TICK-001 (Taxonomy Classification):** Complaints MUST require Category ID, Sub-Category ID, and Complaint Title ID.
- **FR-TICK-002 (Priority & Emergency Handling):** Complaints MUST support priority classifications: `normal` and `emergency`.
- **FR-TICK-003 (Availability Booking):** System MUST record the resident's designated visit window (`availability_time`).
- **FR-TICK-004 (Attachment Support):** System MUST support uploading multiple image attachments per complaint, securely stored with randomized filenames in `/storage/app/public/complaints/`.
- **FR-TICK-005 (12-Stage State Machine):** The `complaints.status` attribute MUST be stored as a `TINYINT` conforming to the exact operational specification:
- `0`: In Progress (`STATUS_IN_PROGRESS`)
- `1`: Addressed / Resolved (`STATUS_RESOLVED`)
- `2`: Unassigned / New (`STATUS_UNASSIGNED` - Default)
- `3`: Assigned (`STATUS_ASSIGNED`)
- `4`: Work Performa (`STATUS_WORK_PERFORMA`)
- `5`: Maintenance Performa (`STATUS_MAINT_PERFORMA`)
- `6`: Work Performa Priced (`STATUS_WORK_PRICED_PERFORMA`)
- `7`: Maintenance Performa Priced (`STATUS_MAINT_PRICED_PERFORMA`)
- `8`: Product N/A (`STATUS_PRODUCT_NA`)
- `9`: Un-Authorized (`STATUS_UN_AUTHORIZED`)
- `10`: Barrack Damages (`STATUS_BARRACK_DAMAGES`)
- `11`: Door Lock (`STATUS_DOOR_LOCK`)
- **FR-TICK-006 (Audit Logging):** Every status change, assignment change, or remark entry MUST generate an immutable record in `complaint_logs` with editor identity, old state, new state, and timestamp.
- **FR-TICK-007 (Citizen Feedback):** Resolved complaints MUST prompt for star ratings (1 to 5) and textual remarks in `complaint_feedbacks`.

3.6 Module 6: Spare Parts & Approval Performa Engine (FR-SPARE)

- **FR-SPARE-001 (Catalog Management):** System MUST track spares with name, item code, brand ID, vendor ID, unit of measure, unit price, and stock on hand.
- **FR-SPARE-002 (Stock Movements):** All incoming stock purchases MUST be logged in `stock_add_data` and historical adjustments in `spare_stock_logs`.

- **FR-SPARE-003 (Performa Creation):** Supervisors MUST be able to generate formal approval performas (`spare_approval_performa`) containing multiple line items (`spare_approval_items`).
- **FR-SPARE-004 (Multi-Level Authorization):** Performas MUST support states: `pending` , `approved` , `rejected` , and `barrack_damages` .
- **FR-SPARE-005 (Automatic Inventory Deduction):** Upon formal approval of a performa, the system MUST automatically decrement the physical quantity from `spares.current_stock` and record issuance in `complaint_spares` .

3.7 Module 7: Service Level Agreements & Escalations (FR-SLA)

- **FR-SLA-001 (Rule Definition):** System MUST allow administrators to define SLA rules based on Category, Priority, and target resolution hours.
- **FR-SLA-002 (Breach Tracking):** The system MUST calculate breach time (`created_at + sla_hours`) and visually highlight overdue tickets on supervisor dashboards.

3.8 Module 8: Mobile RESTful API Gateway (FR-API)

- **FR-API-001 (Payload Format):** All API responses MUST be serialized in standard RFC 8259 JSON format with structured `status` , `message` , and `data` keys.
- **FR-API-002 (Throttling Matrix):**
- `/api/house/login` : Max 5 requests/min.
- `/api/complaints/register` : Max 15 requests/min.
- General read routes: Max 60 requests/min.
- **FR-API-003 (App Version Control):** The endpoint `/api/app/check-update` MUST return mandatory update flags, download URLs, and minimum supported version numbers.

3.9 Module 9: Theme & Accessibility Engine (FR-THEME)

- **FR-THEME-001 (Tri-Mode Switching):** The interface MUST support `Light` , `Dark` , `Night` (low blue-light amber tint), and `Auto` modes.
- **FR-THEME-002 (Persistence):** User theme preferences MUST persist to the database for authenticated users and via HTTP cookies for guests.
- **FR-THEME-003 (WCAG Compliance):** All themes MUST maintain WCAG AA compliant color contrast ratios (minimum 4.5:1 for normal text).

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4. Non-Functional Requirements (NFR)

4.1 Performance & Scalability (NFR-PERF)

- **NFR-PERF-001 (Response Time):** 95% of standard administrative HTTP requests MUST render in under 800 milliseconds on a standard broadband connection.
- **NFR-PERF-002 (API Latency):** Mobile API endpoints MUST respond within 350 milliseconds under a concurrent load of 150 requests.
- **NFR-PERF-003 (Database Indexing):** All foreign key lookup columns and filtering combinations (`city_id` , `sector_id` , `status` , `created_at`) MUST be covered by compound B-tree indexes.

4.2 Security & Data Protection (NFR-SEC)

- **NFR-SEC-001 (SQL Injection Immunity):** All database interactions MUST utilize PDO parameterized queries via Laravel Eloquent ORM. Raw SQL concatenations are strictly forbidden.
- **NFR-SEC-002 (Cross-Site Scripting - XSS):** All Blade templating output MUST use automatic escaping syntax (`{{ $data }}`).
- **NFR-SEC-003 (CSRF Protection):** All incoming POST/PUT/DELETE web requests MUST carry a valid CSRF token.
- **NFR-SEC-004 (File Upload Validation):** Complaint attachments MUST be strictly validated against MIME types (image/jpeg, image/png, image/webp) with a maximum file size ceiling of 5MB. Executable extensions (`.php` , `.sh` , `.exe`) MUST be rejected at the web server and application layers.

4.3 Availability & Reliability (NFR-REL)

- **NFR-REL-001 (Uptime Target):** The system MUST achieve 99.9% uptime, excluding planned maintenance windows.
- **NFR-REL-002 (Soft Deletes):** Core business tables (`complaints` , `spares` , `users` , `employees` , `houses`) MUST implement soft deletion (`deleted_at`) to prevent irreversible accidental data loss.

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5. System Verification & Requirements Traceability Matrix

Requirement ID	Module / Feature	Code Implementation File	Verification Method
`FR-AUTH-001`	Multi-Guard Auth	`app/Http/Controllers/Api/HouseAuthController.php`	Integration Test / Postman
`FR-RBAC-001`	Isolated Exact Permissions	`app/Models/Role.php:hasPermission`	Unit Test
`FR-RBAC-003`	Middleware Route Protection	`routes/web.php` & `app/Http/Middleware/CheckRolePermission.php`	Functional Route Audit
`FR-TICK-005`	12-Stage State Machine	`app/Models/Complaint.php` & `STATUS_MAPPING_REFERENCE.md`	Database Schema / Form Testing
`FR-SPARE-004`	Performa Approval Engine	`app/Models/SpareApprovalPerforma.php`	Workflow Inspection
`FR-SPARE-005`	Automatic Stock Deduction	`app/Http/Controllers/SpareController.php`	Transaction Test
`FR-API-002`	API Rate Throttling	`routes/api.php`	Load Test (ab / wrk)

Requirement ID	Module / Feature	Code Implementation File	Verification Method
`FR-THEME-001`	Tri-Mode UI Accessibility	`app/Http/Controllers/ThemeController.php`	UI Contrast Audit

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